

Tianjian (James) Zhang

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EDUCATION

Columbia University

New York, NY

MS in Electrical Engineering

Feb 2026

GPA: 3.76

Courses: IoT, Data Center Processing, Control, Optimization, Reinforcement Learning, Smart Grid Technology, Robotics, Power Electronics

University of California, Davis

Davis, CA

BS in Physics and Chemical Physics

Jun 2024

Honors: Magna Cum Laude, Outstanding Graduating Senior Awards, Phi Beta Kappa, Dean's List

Courses: Analog Circuits, Digital Circuits, Data Acquisition, Machine Learning, Computer Vision.

EXPERIENCE

Columbia Robotics Club

New York, NY

Electronics Lead

Sep 2024 — Present

- Designed multi-layer PCBs integrating microcontrollers, SPI/I2C protocols, sensors, CAN, and high-speed communication interfaces, ensuring robust board-level functionality.
- Applied signal integrity (SI) principles in PCB board, including controlled impedance routing, grounding strategies, trace length matching, and noise mitigation for reliable high-speed performance.
- Collected and analyzed SI data using oscilloscopes and signal measurement best practices to validate timing margins and minimize crosstalk and reflections.
- Led hardware bring-up and testing, resolving interface and power integrity issues during integration.

Columbia University, DitecT Lab

New York, NY

Research Assistant

Jan 2025 — Dec 2025

- Developed Python-based validation and regression testing tools for camera perception outputs and BEV image processing pipelines on a scaled autonomous vehicle platform.
- Integrated camera modules, IMU, and compute on Nvidia Jetson (Linux), debugging hardware & software synchronization, sensor latency, and pipeline performance issues.

Boldyn Networks

New York, NY

Device Technology Intern

Jun 2025 — Aug 2025

- Supported system-level validation of LTE/5G edge devices, working extensively with Ethernet interfaces and IP-based communication protocols.
- Configured and validated high-speed Ethernet connectivity, analyzing packet loss, throughput, and link stability using network analyzers and diagnostic tools.
- Executed structured regression and certification tests, troubleshooting hardware/software integration and communication timing issues.

Hackathon Projects

New York, NY

Hardware Prototyping

Nov 2024 – Jan 2026

- Rapidly prototyped an embedded posture monitoring system integrating IMU and camera.
- Developed Python-based data processing scripts to analyze and visualize real-time sensor outputs and validate system performance under dynamic conditions with wireless transmission.
- Performed signal debugging, and system integration testing; awarded 3rd place among competing teams.

SKILLS

Programming:

Python, C/C++, Matlab, Bash, Docker, Linux, HTML, SQL, Git, OOP, $\text{\LaTeX}$

Hardware:

FPGA, Jetson, Simulink, Altium, Raspberry Pi, ROS2, Motor Driver

Equipments:

Oscilloscope, Spectrum/Protocol Analyzers, PCB prototyping, Signal Analysis

Other:

Sensor validation, Documentation, Data Analysis, CAD, Jira, Communication